ENGINEERING CALCULATION NOTE / REVISION C

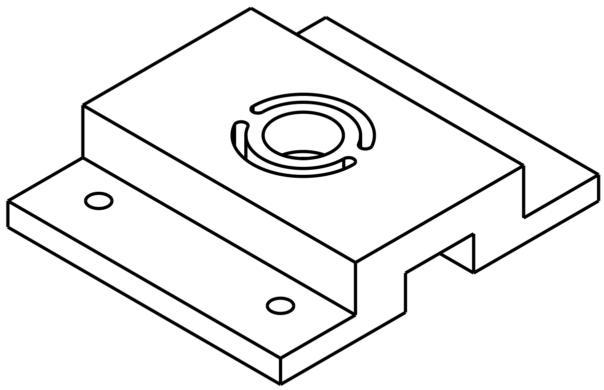
Pigjaw push-out fixture

Can printed PLA replace machined metal?

Hand calculations, retrospective PLA FEA, and model comparison

Prepared for Jerich Lee Units: N, mm, MPa

Answer under the chosen screening assumptions. This unchanged PLA geometry does **not** meet the selected 50 μm maximum-annular-motion target at an assumed 250 N load. The final supplied FEA predicts **131.37 μm**, or **2.63 times the budget**. This establishes a modeled compliance concern; it does not establish fracture, a safe operating load, or that machined metal is mandatory.



Reference CAD supplied as Timmer_NewPushoutforKershLab_v2 v1.step.
The figure and dimensional audit are retained from Revision A.

Compare the responses at the same assumed force

PLA result at 250 N	Downward displacement
Equivalent 60 mm bridge, hand calculation (page 5)	31.77 μm
Final M2 FEA, maximum on loaded annulus (page 10)	131.37 μm
Analyst-selected maximum-annular-motion screen	50.00 μm

At the same force, the bridge value is **75.82% below** the FE maximum. This is a difference between models, not an experimentally measured error. The separate 100 mm-span sensitivity gives 0.13123 mm, close in magnitude but without validating that support span. Aluminum and steel results remain **hand calculations only**.

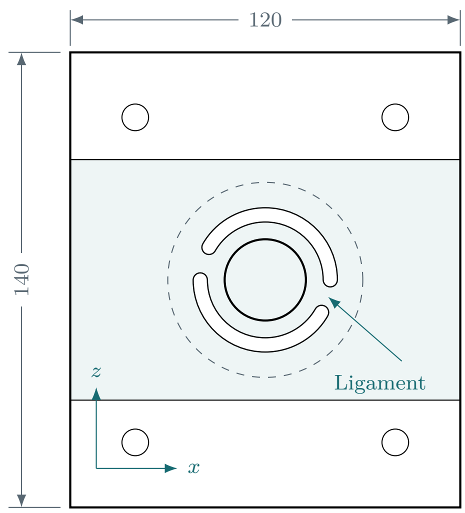
Keep the project history separate from the later analysis

In the original 2025 project, the team did not use this concept because it feared PLA breakage. No observed fracture is established. The hand calculations and supplied FEA were added retrospectively on September 6–7, 2026; they did not drive the earlier decision.

01 / GEOMETRY

Start with the actual load-bearing features

The STEP contains one solid. Its model coordinates have y through the thickness; the assumed test force acts downward along $-y$. The modeled solid is a reference for this calculation, not verified as-built geometry.[1]



Plan view, dimensioned in mm. Dashed circle denotes the 60 mm underside relief; slot linework is schematic.

CAD-derived feature	Value
Overall $x \times z \times y$ envelope	$120 \times 140 \times 30$
Raised deck extent in z	74
Mounting flange thickness	10
Central web, $h = 30 - 15$	15
Central opening, D_h	25
Underside circular relief	60 dia.
Underside cross-channel width	30
Mounting holes	4×8.2 dia.
Mounting pattern in $x \times z$	80×100
Slot inner / outer radii	17.85/22.15
Slot width / end radius	4.30/2.15
Central annulus radial width	5.35

The slot-end gap gives a local minimum width

Adjacent endcap centers in the x–z plane are $(17.3205, -10.0000)$ and $(20.0000, 0.0265)$ mm. Their center distance minus the two end radii gives

$$\begin{aligned} b_\ell &= \sqrt{(20.0000 - 17.3205)^2 + (0.0265 + 10.0000)^2} - 2(2.15) \\ &\simeq \boxed{6.0784\,\mathrm{mm}}. \end{aligned} \tag{1}$$

The calculation uses this minimum width as an equivalent rectangular ligament width. It is not a complete description of the curved load path.

Assumed load path: pusher → implant → interface / surrounding bone → central annulus → slot ligaments → outer deck → rigid base. Contact on the outer deck can bypass part of the annulus load; that possible load sharing is not quantified by this model.

> **Do not use the 30 mm overall height as the bending thickness.** The central material above the underside relief is only 15 mm thick. For the same rectangular beam, substituting 30 mm would overpredict bending stiffness by $(30/15)^3 = 8$.

02 / LOAD AND FIT

Define the force envelope and exit clearance

What the supplied papers establish

Dewey et al. describe a critical-size porcine scaffold geometry of **25 mm diameter and 10 mm height**; their study is not a source of the present campaign's peak push-out force.[2] Lawson et al. validated a different, **80%-infill ABS** fixture only at loads ≤ 250 **N**, using smaller rat/rabbit calvarial specimens. That result is neither a PLA qualification nor a porcine force limit.[3]

Accordingly, this note uses $F = 250\,\mathrm{N}$ as a literature-context benchmark and $F = 500$ and $1000\,\mathrm{N}$ as explicitly assumed sensitivity cases. The actual required force F_{req} remains an input to be established.

A useful force conversion, not a strength prediction

For an idealized cylindrical bonded interface, its nominal lateral area is

$$A_i = \pi D_i t_i, \qquad \tau_{\mathrm{app}} = \frac{F}{\pi D_i t_i}, \qquad F = \tau_{\mathrm{app}} \pi D_i t_i. \tag{2}$$

Here t_i is the *effective axial contact length*, not fixture thickness h, and not automatically the scaffold's full height. This nominal stress follows the conventional push-out definition; local interface stress need not be uniform.[4]

Assumed F	τ_{app} for $D_i = 25,\ t_i = 10$ mm	τ_{app} for $D_i = 25,\ t_i = 3$ mm
250 N	0.318 MPa	1.061 MPa
500 N	0.637 MPa	2.122 MPa
1000 N	1.273 MPa	4.244 MPa

The 3 mm and 10 mm contact lengths are sensitivity assumptions here. Do not import coated-metal implant strengths or Teflon press-fit retention forces as healed porcine scaffold strengths; Dewey 2019's Teflon-hole test addresses a different interface.[5]

The 25 mm opening requires an independent clearance check

$$c = \frac{D_h - D_i}{2}. \tag{3}$$

For $D_h = D_i = 25\,\mathrm{mm}$, $c = 0$. Dhert's model recommended a radial clearance of at least $0.7\,\mathrm{mm}$; Lawson adopted that starting point for smaller specimens.[4, 3] Applying it provisionally gives

$$D_h \geq D_i + 2c = 25 + 2(0.7) = \boxed{26.4\,\mathrm{mm}}, \tag{4}$$

before tolerances and alignment allowance. With a 25 mm hole, the corresponding implant diameter would be at most $23.6\,\mathrm{mm}$.

> **Conditional finding, not a universal design rule.** The actual implant diameter and release path must be measured. The 0.7 mm recommendation was not established for this 25 mm porous porcine construct. Enlarging the hole also narrows the annulus, so repeat the structural checks after any change.

03 / INPUTS

Separate source properties from design assumptions

Trial materials for the same geometry

Material	E (MPa)	ν	σ_{allow} (MPa)	τ_{allow} (MPa)
Printed PLA surrogate	2000	0.35	10	5
6061-T6/T651 aluminum	68300	0.33	120	60
304 stainless, EN plate basis	200000	0.30	105	52.5

PLA: UltiMaker's printed, 100%-infill coupons have flexural moduli of approximately 2.74–3.02 GPa and upright tensile break strength 33.1 ± 2.8 MPa, with no tensile yield reported for that orientation.[6] Here $E = 2.0$ GPa and the 10/5 MPa normal/shear allowables are **analyst-selected screening assumptions**, not measurements of your print and not statistical lower bounds. The normal allowance corresponds to an assumed 20 MPa reference strength divided by two. The shear allowance is a separate conservative trial value, not a validated PLA failure law.

Aluminum: $E = 68.3$ GPa follows Kaiser's typical 6061 data.[7] A reference yield strength of 240 MPa is rounded down from the 35 ksi plate limit listed for the applicable thickness range; the trial normal allowance is $240/2 = 120$ MPa.[8]

Steel: Outokumpu lists $E = 200$ GPa and 210 MPa proof strength for Core 304 plate on its EN basis; $210/2 = 105$ MPa is used here.[9] This is not a generic guarantee for every material sold as "304." Confirm the purchasing specification and material certificate.

The metal shear allowances are half the normal allowances, a conservative pure-shear, Tresca-based screen. The Poisson ratios in the table are nominal modeling assumptions. For calculations,

$$G = \frac{E}{2(1+\nu)}, \qquad G_{\text{PLA}} = \frac{2000}{2(1.35)} = 740.7\,\text{MPa}. \tag{5}$$

For printed PLA this isotropic relationship is only a surrogate; interlayer shear stiffness may differ substantially.

Explicit acceptance targets for this exercise

$$\boxed{\delta_{\text{fixture}} \leq 0.05\,\text{mm} = 50\,\mu\text{m}}, \qquad \sigma_{\text{screen}} \leq \sigma_{\text{allow}}, \qquad \tau_{\text{screen}} \leq \tau_{\text{allow}}. \tag{6}$$

The 50 μm displacement budget is chosen for this note, not imposed by the papers. This report applies it to the maximum downward movement on the FE loaded annulus. This screening metric is not automatically specimen-displacement error (page 10). The strength factor of two is already included in the trial allowables: do not apply it a second time to the same comparison.

> All numerical checks assume short-duration, quasi-static loading and the adopted effective material properties. Unknown infill, print direction, defects, wet conditioning, temperature, creep, and load dwell prevent a final PLA qualification. The supplier data were obtained under stated room-temperature conditioning, not your complete laboratory exposure.[6]

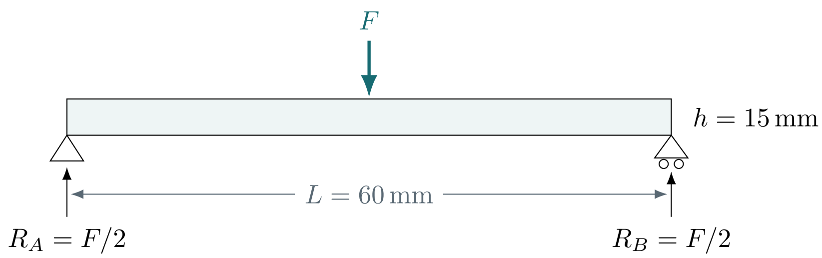
04 / GLOBAL MODEL

Replace the deck by an equivalent bridge

Assume the bottom lands are backed by a rigid base, and take effective simple supports at $z = \pm 30\,\text{mm}$, beyond the circular relief. Use $L = 60\,\text{mm}$ and $h = 15\,\text{mm}$. Remove the entire 44.3 mm-diameter slot envelope from the width, over the full span:

$$b_g = 120 - 2(22.15) = 75.7\,\text{mm}, \quad A_g = b_g h = 1135.5\,\text{mm}^2, \quad I_g = \frac{b_g h^3}{12} = 21290.6\,\text{mm}^4. \tag{7}$$

This is a **global bridge surrogate**, not a literal beam present in the CAD. It lumps load spreading into a central resultant and does not resolve the annulus or slot ligaments. Removing material reduces this surrogate's stiffness, but does not make it a rigorous upper bound on the real fixture's deflection.



1. Reactions, bending moment, and nominal stress

$$R_A = R_B = \frac{F}{2}, \qquad M_{\max} = \frac{FL}{4},$$
$$\sigma_{g,\max} = \frac{M_{\max}(h/2)}{I_g} = \frac{3FL}{2b_g h^2}. \tag{8}$$

At $F = 250\,\text{N}$: $R_A = R_B = 125\,\text{N}$, $M_{\max} = 3750\,\text{N mm}$, and

$$\sigma_{g,\max} = \frac{3(250)(60)}{2(75.7)(15)^2} = \boxed{1.321\,\text{MPa}}.$$

2. Bending plus transverse-shear deflection

Integrating the beam curvature with simple-support conditions gives $\delta_b = FL^3/(48EI_g)$. The shear-energy term gives $\delta_s = FL/(4\kappa G A_g)$, with the rectangular-section surrogate $\kappa = 5/6$:

$$\boxed{\delta_g = \frac{FL^3}{48EI_g} + \frac{FL}{4\kappa G A_g}}. \tag{9}$$

For PLA at 250 N,

$$\delta_b = \frac{250(60)^3}{48(2000)(21290.6)} = 0.02642\,\text{mm},$$
$$\delta_s = \frac{250(60)}{4(5/6)(740.7)(1135.5)} = 0.00535\,\text{mm},$$
$$\delta_g = \boxed{0.03177\,\text{mm} = 31.77\,\mu\text{m}}. \tag{10}$$

Shear contributes about 17% of this total; omitting it is not ideal for the short $L/h = 4$ bridge. Passing this check alone does not pass the slotted fixture.

05 / LOCAL CHECKS

Check the annulus and its two narrow connections

Assume, for screening, that the entire test reaction enters the central annulus and then divides equally between its two ligaments. Actual specimen contact can instead share load with the outer deck; eccentric contact can also overload one ligament. The load partition must be checked physically.

1. Average contact pressure: a necessary but weak check

Taking the annulus from $r_i = 12.5$ to $r_o = 17.85\,\text{mm}$ gives

$$A_c = \pi(r_o^2 - r_i^2) = \pi(17.85^2 - 12.5^2) = 510.1\,\text{mm}^2,$$
$$p_{\text{avg}} = \frac{F}{A_c} = \frac{250}{510.1} = \boxed{0.490\,\text{MPa}}. \tag{11}$$

An optimistic uniform-column shortening estimate is

$$\delta_c = \frac{Fh}{EA_c} = \frac{250(15)}{2000(510.1)} = 0.00368\,\text{mm}. \tag{12}$$

This excludes bending and assumes full-area contact. If only a fraction η contacts, replace A_c by ηA_c. Neither p_{avg} nor δ_c establishes specimen safety or fixture adequacy.

2. Ligament shear: much less area carries the reaction

With $b_\ell = 6.0784\,\text{mm}$, $h = 15\,\text{mm}$, and $V = F/2$ per ligament,

$$\tau_{\ell,\text{avg}} = \frac{F}{2b_\ell h} = \frac{250}{2(6.0784)(15)} = 1.371\,\text{MPa}. \tag{13}$$

For a rectangular-section peak factor of 1.5 and an *assumed* local concentration multiplier $K_s = 2$,

$$\tau_{\ell,\text{screen}} = K_s\frac{3F}{4b_\ell h} = \boxed{4.113\,\text{MPa} \quad \text{at } 250\,\text{N}}. \tag{14}$$

The ratio to the PLA trial allowance is $4.113/5 = 0.823$. A shear-only crossover occurs at $F \approx 304\,\text{N}$, but this is **not an assembly load rating**. K_s is a sensitivity choice, not a tabulated factor for this CAD.

3. Local bending needs an effective moment arm

Represent each connection by a rectangular cantilever with resultant arm ℓ. Then $M_\ell = (F/2)\ell$ and, for an *assumed* $K_t = 2$,

$$\sigma_{\ell,\text{screen}} = K_t\frac{3F\ell}{b_\ell h^2}. \tag{15}$$

Assumed ℓ	$\sigma_{\ell,\text{screen}}$ at 250 N	Against PLA's 10 MPa allowance
5 mm	5.48 MPa	Below allowance
10 mm	10.97 MPa	Slightly above allowance
15 mm	16.45 MPa	Above allowance

ℓ is not a measured slot length. It represents the unknown contact/resultant and restraint arrangement. The annulus can bend and twist, and the ligament roots are not ideal cantilever clamps. This sensitivity prevents a falsely precise PLA "pass."

06 / COMPARISON

The material gain is clear; the absolute rating is not

Global bridge results for the declared 60 mm support span

F (N)	$\sigma_{g,\max}$ (MPa)	δ_g in μm		
		PLA	Aluminum	Stainless
100	0.528	12.71	0.371	0.126
250	1.321	31.77	0.928	0.316
500	2.642	63.54	1.856	0.631
1000	5.284	127.08	3.712	1.263

These are linear-elastic *global-model* estimates. They omit local ring motion and joints; higher-load PLA entries are extrapolations, not predictions after local damage. The nominal stress is the same for all materials in this statically determinate model. The allowables and deflections differ.

$$\frac{(EI)_{\mathrm{Al}}}{(EI)_{\mathrm{PLA}}} = \frac{68300}{2000} = \boxed{34.15}, \qquad \frac{(EI)_{\mathrm{steel}}}{(EI)_{\mathrm{PLA}}} = \frac{200000}{2000} = \boxed{100}. \tag{16}$$

Steel gives about 2.93× aluminum's flexural stiffness at the same geometry. A large stiffness gain does not automatically justify steel if an aluminum design is later verified to satisfy the required accuracy with margin.

Geometry and restraint can change the decision

PLA sensitivity, all at 250 N	Global δ_g	Meaning
Baseline: $E = 2$ GPa, $L = 60$ mm	$31.77\,\mu$m	Uses 64% of 50 μm
Stiffer print: $E = 3$ GPa, same ν	$21.18\,\mu$m	Still excludes local motion
More flexible support: $L = 100$ mm	$131.23\,\mu$m	Exceeds chosen budget

The 100 mm span is an alternate boundary-condition scenario. Although it equals the CAD mounting-row spacing, **bolt spacing does not establish a bearing span**. Actual support contact, not the nominal bolt pattern, determines the appropriate model.

For the baseline PLA bridge, compliance is

$$C_g = \frac{\delta_g}{F} = 1.2708 \times 10^{-4}\,\mathrm{mm/N}, \qquad F_{\delta,g} = \frac{0.05}{C_g} = \boxed{393\,\mathrm{N}}. \tag{17}$$

This 393 N value is only the point where the *global surrogate* uses the entire displacement budget. Local bending, shear, creep, and joints can set a lower usable force.

> **A useful design lever:** $\delta_b \propto L^3/(Ebh^3)$ and $\delta_s \propto L/(Gbh)$. Shortening the unsupported span or thickening the web can make a printed fixture more viable. Removing or repositioning slots can help only after confirming their assembly/alignment function.

07 / TEST QUALITY

Survival is not the same as measurement fidelity

Fixture compliance biases crosshead-based stiffness

For an ideal series load path, the same force passes through the specimen and fixture, but displacement is shared:

$$u_{\text{crosshead}} = u_s + FC_{\text{fixture}} + FC_{\text{machine}} + u_{\text{seating}}. \tag{18}$$

In a settled, linear interval, with total parasitic compliance C_p,

$$\frac{1}{k_{\text{measured}}} = \frac{1}{k_s} + C_p, \qquad \frac{k_s - k_{\text{measured}}}{k_s} = \frac{k_s C_p}{1 + k_s C_p}. \tag{19}$$

A maximum fractional stiffness error ε requires

$$k_p = \frac{1}{C_p} \geq \frac{1-\varepsilon}{\varepsilon} k_s; \qquad \varepsilon = 5\% \implies \boxed{k_p \geq 19 k_s}. \tag{20}$$

Example only: for an assumed $k_s = 500\,\mathrm{N/mm}$, the baseline PLA global bridge alone gives a 5.97% apparent stiffness reduction; the aluminum bridge gives about 0.185%. Other compliance adds to this. Neither k_s nor these errors was measured for the porcine specimens.

Pure elastic compliance does not by itself make an ideal in-line load cell read the wrong static force. It does affect inferred displacement, stiffness, energy, and the specimen's actual displacement rate. Interference, lateral force, slip, and changing contact are separate errors. Lawson specifically emphasizes coaxial alignment and avoiding unintended contact during push-out.[3]

Mounting preload can exceed the test force

The STEP has 8.2 mm mounting holes, not a complete fastener specification. For an *illustrative* M8 fastener only, use the rough torque model $T = KF_p d_b$ with assumed $K = 0.20$:

$$T = 2\,\mathrm{N\,m} = 2000\,\mathrm{N\,mm}, \quad d_b = 8\,\mathrm{mm} \implies F_p \approx \frac{2000}{0.20(8)} = 1250\,\mathrm{N}. \tag{21}$$

For a hypothetical 24 mm OD / 8.2 mm ID washer,

$$A_w = \frac{\pi}{4}(24^2 - 8.2^2) = 399.6\,\mathrm{mm}^2, \qquad p_w \approx \frac{1250}{399.6} = \boxed{3.13\,\mathrm{MPa}}. \tag{22}$$

This is a sensitivity calculation, **not a recommended tightening torque**. Sustained pressure can cause a printed joint to relax. The FEA instead uses **16 mm OD patches with prescribed $Y = 0$ and no preload**; this 24 mm washer example is not an FE input.

> Support the lower lands on a verified rigid base and measure the fixture's own motion. Broad washers or suitable metal sleeves can protect printed mounting regions, but cannot eliminate deck or ligament bending. Pusher, stud, clamp, fastener, and base capacity must be checked separately; they are not contained in this single-solid STEP.

08 / FEA MODEL

A geometry-resolved follow-up to the hand screen

The following results were supplied from ANSYS Mechanical 2025 R2. They are screenshot readouts and recorded setup, not an independent solver rerun. The single fixture body uses an **isotropic linear-elastic PLA surrogate**, $E = 2000\,\mathrm{MPa}$ and $\nu = 0.35$, matching the hand-calculation inputs.[11]

Load and modeled contact

Uniform pressure acts downward ($-Y$) on the top annulus, ID 25 mm / OD 35.7 mm:

$$\begin{aligned} A_c &= \frac{\pi}{4}(35.7^2 - 25^2) = 510.1083\,\mathrm{mm}^2, \\ F_{\mathrm{ann}} &= pA_c = 0.4901(510.1083) = 250.004\,\mathrm{N} \simeq 250\,\mathrm{N}. \end{aligned} \tag{23}$$

The pressure supplies the whole test load; there is no second 250 N force. This annular-only loading is a **defined scenario**, not verified specimen contact or a demonstrated worst case. Actual contact could share load with the outer deck or load the ring eccentrically.

Boundary condition	Recorded representation
Plate backing	Compression Only Support on the two lowest flat bottom lands; recess, channel ceiling, and hole/slot walls unsupported. Sliding friction neglected.
Contact control	Manual normal stiffness factor 10; update each iteration. This is a numerical contact factor, not a plate modulus.
Washer hold-down	Four flange-top annular patches, assumed OD 16 mm / ID 8.2 mm, total area about 593 mm^2. $Y = 0$; X, Z free. No washer solid or bolt pretension.
Locate A	One bottom outer corner: $X = 0$, $Z = 0$, Y free.
Locate B	Other endpoint of the same X-directed edge: $Z = 0$, X, Y free. The points share Y, Z and differ in X.

The user's clarification that bolts and washers held the flanges to a flat rigid surface superseded the initial simply-resting assumption. The restraints approximate vertical hold-down and in-plane location; they do not reproduce real preload, washer bending, bolt stretch, or friction.

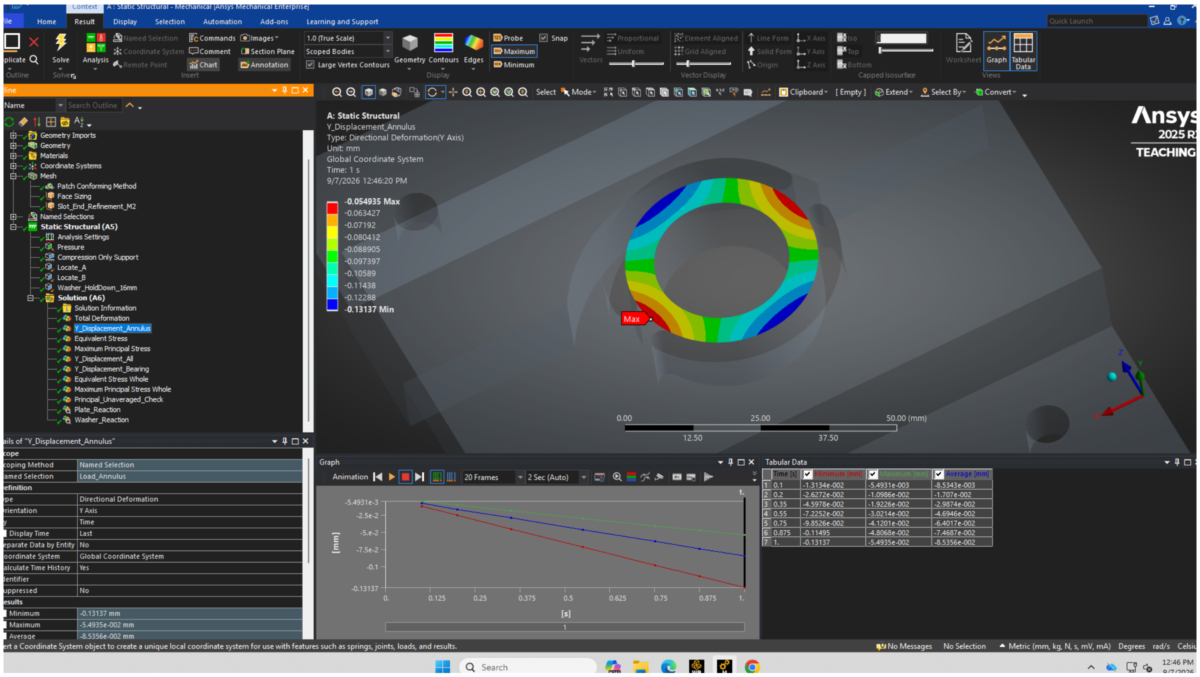
Analysis controls and reproducibility

The body is flexible; large deflection is off; no gravity is applied. The recorded setup uses one static load step, end time 1 s, automatic substeps (initial 10, minimum 5, maximum 100), program-controlled solver, and weak springs off. **Time = 1 s is the final static load-step parameter**, not a physical loading rate or dwell. Compression-only contact can open and close despite the linear material model.

> The supplied STEP predates the annular and washer face partitions. No final ANSYS project, solver deck, result file, or full solver log was supplied. Some setup details are recorded workflow settings rather than independently audited inputs. Exact reproduction requires the final project or reconstruction of its partitions and settings.

09 / DISPLACEMENT

Final PLA model exceeds the chosen motion budget



Supplied M2 screenshot, unmodified. Global Y directional deformation scoped to the loaded annulus; units mm; final static step. Displayed shape is visualization.

Read the signed extrema correctly

The final annular minimum is $u_{Y,\text{min}} = -0.13137\,\text{mm}$; the displayed maximum is $u_{Y,\text{max}} = -0.054935\,\text{mm}$. Both are downward. The **most negative** value gives the largest downward movement:

$$\begin{aligned}
\delta_{\text{ann,max}} &= |u_{Y,\text{min}}| = \boxed{0.13137\,\text{mm} = 131.37\,\mu\text{m}},\\
\frac{\delta_{\text{ann,max}}}{\delta_{\text{budget}}} &= \frac{0.13137}{0.05000} = \boxed{2.6274},\\
\delta_{\text{ann,max}} - \delta_{\text{budget}} &= 0.08137\,\text{mm} = \boxed{81.37\,\mu\text{m}}.
\end{aligned} \tag{24}$$

The difference between annular extrema is $0.076435\,\text{mm} = 76.435\,\mu\text{m}$. It shows spatially nonuniform fixture movement; it does not by itself establish rigid-specimen tilt or interface deformation.

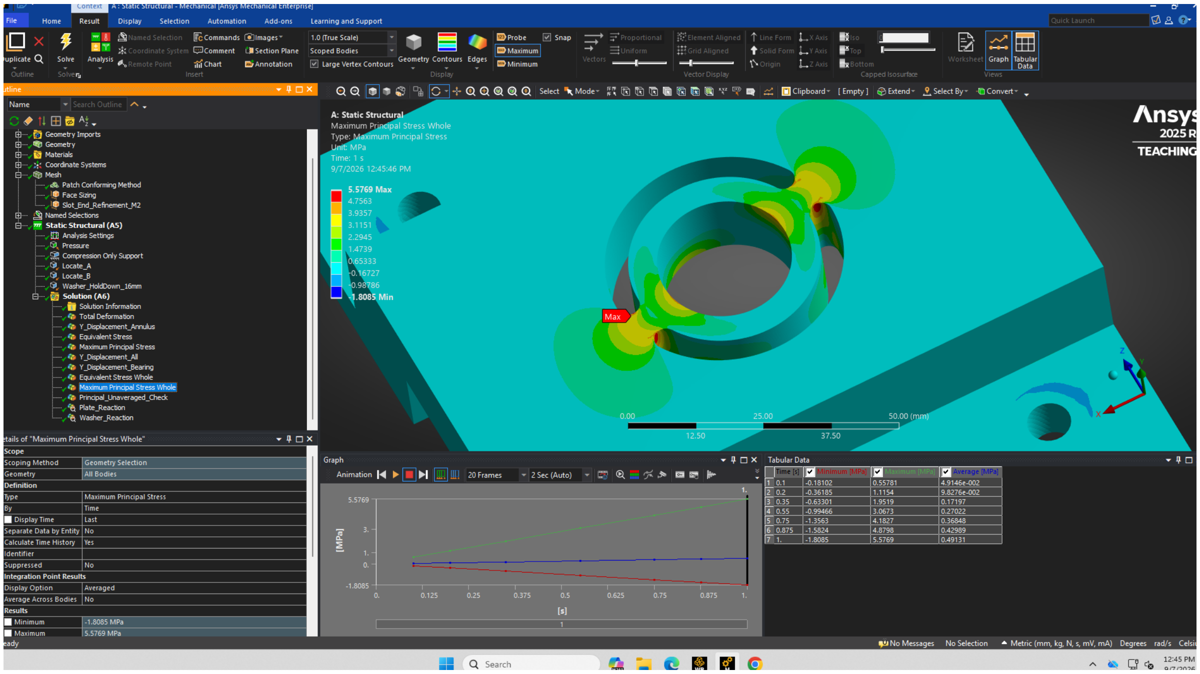
> **Screening outcome: does not meet the chosen target.** At the assumed 250 N benchmark, this PLA model exceeds the 50 μm maximum-annular-motion budget. The final local mesh refinement changed this maximum by only 0.03 μm (page 12), which is small relative to the 81.37 μm exceedance.

What this displacement can and cannot mean

These are local fixture responses. Neither an extremum nor the mesh-dependent nodal arithmetic average is automatically specimen displacement, crosshead error, or a work-conjugate compliance. Do not use either as a scalar correction in the page 8 equations. Relating fixture motion to measurement accuracy requires the actual specimen contact and measurement arrangement, which are absent from this model.

10 / LOCAL STRESS

Treat the tensile peaks as provisional indicators



Supplied M2 screenshot, unmodified. Maximum principal stress over the whole fixture, averaged display, units MPa; final static step.

Maximum principal tensile stress	M1	M2
Averaged peak (MPa)	5.4035	5.5769
Unaveraged peak (MPa)	6.4151	6.0484

Averaging changes the reported peak

Using the two final reported maxima,

$$\frac{\sigma_{1,\max}^{\text{unavg}} - \sigma_{1,\max}^{\text{avg}}}{\sigma_{1,\max}^{\text{avg}}}\,100 = \frac{6.0484 - 5.5769}{5.5769}\,100 = \boxed{8.45\%}. \tag{25}$$

The maxima can occur at different locations; this gap is not a local error estimate. These are two postprocessing representations of one FE solution, not an uncertainty interval enclosing the true stress.

> **No complete strength pass or fracture conclusion.** Both tensile readouts are below the assumed 10 MPa normal allowance, but that comparison does not qualify a printed PLA part. Final von Mises, maximum shear, and minimum principal stress results were not supplied. Anisotropy, interlayer weakness, creep, damage, and cracking are absent from the model; the trial allowance is not a measured failure criterion.

The averaged peak changes by +3.21% from M1 to M2, while the unaveraged peak changes by −5.72%. The latter misses the chosen 5% successive-change criterion. In addition, the overlap of the lowest-quality element with the stress maximum is unresolved (page 12). **Local strength assessment therefore remains provisional**, even though displacement is stable in the final refinement.

11 / NUMERICAL CHECKS

Stable displacement; incomplete stress convergence

All three levels use quadratic tetrahedra, Patch Conforming meshing, and adaptive sizing off. M0–M2 share the idealized bolted restraint and factor-10 contact control. The earlier unclamped run is a separate boundary-condition case.[11]

Mesh	Sizing controls	Nodes	Elements
M0	4 mm global; 1 mm annular face	52,868	32,573
M1	2 mm global; 1 mm annulus with 20 mm influence distance	891,963	629,683
M2	Same as M1; four rounded slot-end walls at 0.5 mm with 4 mm influence distance	1,321,643	941,817

M2 is a **local 0.5 mm slot-end refinement**, not a global 0.5 mm mesh.

Reported response	M0	M1	M2
Maximum downward annular motion (mm)	0.12942	0.13134	0.13137
Least downward annular motion (mm)	0.053713	0.054926	0.054935
Averaged principal tensile peak (MPa)	4.6608	5.4035	5.5769
Unaveraged principal tensile peak (MPa)	Not supplied	6.4151	6.0484

Successive changes, relative to the preceding mesh

Refinement	Displacement	Averaged stress	Unaveraged stress
M0 → M1	+1.48%	+15.94%	Not available
M1 → M2	+0.02284%	+3.209%	−5.716%

The final motion change is $0.00003\,\text{mm} = 0.03\,\mu\text{m}$. This is sensitivity to the last local refinement, **not accuracy to 0.023%**. The gap between unaveraged and averaged maxima narrows from 18.72% (M1) to 8.45% (M2); maxima can occur at different locations, so this is not a local error estimate. Percentage definitions appear on page 13.

Contact and element-quality checks

On the coarse bolted mesh, switching from program-controlled contact to manual factor 10 changes maximum downward motion from 0.13024 to 0.12942 mm, a **0.630% decrease**. This is one contact-parameter sensitivity check; it does not establish all contact settings are insensitive.

M1 minimum/mean Element Quality is 0.23645/0.84389, retained from a handoff transcription; its original quality screenshot is not included. M2 is **0.077138/0.84497**. One element in the lowest histogram bin was isolated near a curved boundary; other low-quality bins exist. Coincidence with the stress peak was not established. This limits local stress confidence without, by itself, invalidating the displacement screen.

> **Report precisely:** final annular displacement changed by about 0.023%; averaged peak stress changed by 3.21%; unaveraged peak stress changed by 5.72%. Do not label all stresses "fully converged." Percentages inherit the rounded screenshot precision.

12 / EQUILIBRIUM AND METRICS

Define the reference behind every percentage

The final vertical reactions balance the applied load

The M2 force-reaction screenshots give the following global Y components at the final static step:

Quantity	Signed Y force (N)
Plate reaction, compression-only support	+355.55
Washer reaction, combined over four patches	−105.50
Sum of support reactions	+250.05
Nominal applied annular load	−250.00

$$r_Y = 355.55 - 105.50 - 250.00 = +0.05\,\mathrm{N}, \qquad \frac{|r_Y|}{250}\,100 = \boxed{0.020\%}. \tag{26}$$

Using $pA = 250.004\,\mathrm{N}$ instead gives a residual of approximately 0.046 N (0.018%). Both indicate close vertical balance relative to 250 N. Values derive from rounded screenshot readouts and do not establish solver tolerances. This is a **vertical force-balance check**, not a full moment-balance or solver-log audit.

The downward washer reaction is compatible with the prescribed hold-down preventing local lift. It is the combined reaction of four constrained patches; it is neither an imposed bolt preload nor a per-bolt force. Add signed components, not reaction magnitudes.

Three different questions require different evidence

For hand-calculation (HC) comparisons, use positive displacement magnitudes or the stated stress values, with the final FE readout as the numerical reference:

$$\boxed{e_{\mathrm{HC|FE}} = \frac{q_{\mathrm{HC}} - q_{\mathrm{FE}}}{|q_{\mathrm{FE}}|}\,100\%.} \tag{27}$$

A negative value means the hand value is lower; a positive value means it is higher. For mesh refinement, use the preceding mesh instead:

$$\boxed{\Delta_{\mathrm{mesh}} = \frac{q_{\mathrm{new}} - q_{\mathrm{previous}}}{|q_{\mathrm{previous}}|}\,100\%.} \tag{28}$$

Quantity	What it establishes
Model discrepancy	Difference between stated model outputs. FEA as denominator is a reference choice, not experimental truth.
Numerical sensitivity	Change under a specified mesh/contact variation. It is not a complete numerical error bound.
Experimental accuracy	Requires physical measurements with matching response definitions. None were provided for this concept.

> The 0.020% force residual is not a displacement-accuracy percentage. Similarly, the 0.023% final refinement change does not establish physical accuracy. The comparisons on pages 15–16 use the explicitly printed, rounded values; their response definitions still differ.

13 / MODEL ASSUMPTIONS

The global bridge does not resolve local load transfer

The original note contains several separate screens: a global bridge, local ligament bending/shear, annular compression, clearance, and illustrative mounting preload. They are not interchangeable solutions and must not be automatically added as a complete fixture response.

The bridge arithmetic included transverse shear

For the simply supported equivalent section, $L = 60\,\mathrm{mm}$, $b_g = 120 - 44.3 = 75.7\,\mathrm{mm}$, $h = 15\,\mathrm{mm}$, $A_g = 1135.5\,\mathrm{mm}^2$ and $I_g = 21290.625\,\mathrm{mm}^4$. The supports were assumed at $z = \pm 30\,\mathrm{mm}$; this is not a literal beam in the CAD.

$$
\begin{aligned}
E = 2000\,\mathrm{MPa}, \quad \nu = 0.35, \quad G = \frac{E}{2(1+\nu)} \simeq 740.7\,\mathrm{MPa}, \\
\delta_g = \underbrace{\frac{FL^3}{48EI_g}}_{\text{bending}} + \underbrace{\frac{FL}{4\kappa G A_g}}_{\text{transverse shear}}, \quad \kappa = \frac{5}{6}, \\
\delta_g\big|_{250\,\mathrm{N}} = 0.02642 + 0.00535 = \boxed{0.03177\,\mathrm{mm}}. \qquad (29)
\end{aligned}
$$

Using the displayed components, the shear fraction is $0.00535/0.03177 = \boxed{16.84\%}$. The calculations reproduce the original result. An arithmetic mistake or omission of this shear term does not explain the difference from FEA.

The structural idealizations differ

Feature	Hand-calculation approximation	Final FE representation
Global geometry	Equivalent constant-section bridge	Imported three-dimensional slotted solid
Load introduction	Central resultant	Pressure distributed over the annulus
Local deformation	Ring and connections absent from the global bridge	Connected ring/ligament geometry included
Bottom support	Effective simple supports at chosen span	Compression-only contact on bottom bearing faces
Hold-downs	Absent from global bridge	$Y = 0$ on four assumed 16 mm washer footprints
Ligament loading	Separate local check assumes $F/2$ per connection	Distribution follows modeled equilibrium and deformation compatibility
Stress concentration	Trial $K_t = K_s = 2$ in local screens	Explicit geometry; no extra multiplier applied to FE stress
Reported motion	Beam midspan displacement	Maximum downward nodal motion anywhere on loaded annulus

Both approaches share assumed PLA properties and an assumed 250 N benchmark. The FEA remains an idealization: it does not resolve actual preload, washer flexibility, sliding friction or print architecture. Agreement between models cannot validate assumptions they share.

14 / DISPLACEMENT COMPARISON

The 60 mm bridge value is 75.82% below the FE maximum

At the matched 250 N benchmark, use $\delta_{\mathrm{FE,max}} = 0.13137\,\mathrm{mm}$ as the comparison reference. Values below are the original report's rounded hand results, compared with the final rounded M2 readout.

Hand model at 250 N	Motion (mm)	$e_{\mathrm{HC\|FE}}$
Uniform-column compression only	0.00368	−97.20%
60 mm bridge, bending term only	0.02642	−79.89%
60 mm bridge, bending plus shear	0.03177	−75.82%
Alternative 100 mm bridge, bending plus shear	0.13123	−0.11%

The first two rows are deliberately incomplete descriptions, not competing complete solutions. The bridge predicts beam midspan motion; the FE quantity is an annular nodal maximum. These are screening comparisons, not measured prediction errors.

The same difference, with two denominators

$$
\begin{aligned}
\delta_{\mathrm{FE,max}} - \delta_{60} &= 0.13137 - 0.03177 = \boxed{0.09960\,\mathrm{mm} = 99.60\,\mu\mathrm{m}},\\
e_{\mathrm{HC|FE}} &= \frac{0.03177 - 0.13137}{0.13137}\,100\% = \boxed{-75.82\%},\\
\frac{\delta_{\mathrm{FE,max}} - \delta_{60}}{\delta_{60}}\,100\% &= \frac{0.13137 - 0.03177}{0.03177}\,100\% = \boxed{313.50\%}.
\end{aligned}
\tag{30}
$$

Thus the hand value is 75.82% below the FE maximum, while the FE maximum is 313.50% above the hand value, or **4.135 times** as large. The denominator must accompany the percentage.

Why a small compression estimate was insufficient

The $Fh/(EA_c)$ estimate assumes a uniformly compressed column over the annular area. The relief and slots instead redirect the fixture's load through the surrounding structure. The approximately 97% difference illustrates the limitation of a compression-only screen; a low average contact pressure did not establish fixture adequacy.

The 100 mm case was a prior sensitivity, not a later fit

The original hand report already contained the 100 mm-span case. Its printed result differs from M2 by only

$$
0.13137 - 0.13123 = \boxed{0.00014\,\mathrm{mm} = 0.14\,\mu\mathrm{m}}. \tag{31}
$$

This prior sensitivity reached the same displacement scale as FEA. However, mounting-row spacing does not establish bearing span, and matching one scalar does not establish the load path or deformation mechanism. Neither the 100 mm span nor its simple-support model is validated by this numerical proximity.

> The separate **0.12708 mm at 1000 N** bridge result must not be compared with **0.13137 mm at 250 N** as same-load agreement. Metal scenarios remain hand calculations; no aluminum or steel FE comparison exists.

15 / STRESS COMPARISON

The local checks usefully identified regions to investigate

Stress comparisons require matching definitions. The hand models estimate nominal or locally scaled bending normal stress; M2 reports principal stress from a three-dimensional stress state. Its averaged and unaveraged peaks are different displays of one solution, not bounds on the true stress.

Global nominal bending is not a local peak prediction

At 250 N the 60 mm bridge gives $M_{\max} = 3750\,\mathrm{N\,mm}$ and $\sigma_{g,\max} = 1.321\,\mathrm{MPa}$. This is numerically **76.31% below** the FE averaged principal peak of 5.5769 MPa. No corresponding nominal FE section stress was extracted, so this is not a matched stress-component error. It shows why a nominal bridge stress could not substitute for a local peak.

Local ligament bending was more targeted

The local screen assumes two equal $F/2$ load paths, width $b_\ell = 6.0784\,\mathrm{mm}$, thickness $h = 15\,\mathrm{mm}$, an effective cantilever arm ℓ and trial concentration factor $K_t = 2$:

$$M_\ell = \frac{F\ell}{2}, \qquad \boxed{\sigma_{\ell,\mathrm{screen}} = K_t \frac{3F\ell}{b_\ell h^2}.} \tag{32}$$

Using the **printed rounded hand stresses**, compare with the M2 averaged peak (5.5769 MPa) and unaveraged peak (6.0484 MPa):

Assumed ℓ	Hand stress	Versus FE averaged	Versus FE unaveraged
5 mm	5.48 MPa	−1.74%	−9.40%
10 mm	10.97 MPa	+96.70%	+81.37%
15 mm	16.45 MPa	+194.97%	+171.97%

Each percentage uses its column's FE peak as denominator. These are **screening-value discrepancies across different stress definitions**, not strict stress-prediction errors. Retaining the rounded hand inputs matters; using their unrounded source values changes the last percentage digits.

The 5 mm-arm case is close in magnitude. The 10 and 15 mm cases are higher than the displayed FE tensile peaks for this load case, but that does not establish conservative bounds for other stresses or failure modes. The supplied plots show elevated tensile stress near the narrow annulus connections and rounded slot ends, consistent with the hand checks' qualitative focus.

> Neither $\ell = 5\,\mathrm{mm}$ nor $K_t = 2$ is independently validated by a close scalar result. The actual moment distribution and local stress-concentration factor were not extracted separately. Qualitative hotspot agreement is not exact location or parameter validation.

No matching final FE shear result was supplied

The separate hand check gives $\tau_{\ell,\mathrm{avg}} = 1.371\,\mathrm{MPa}$ and $\tau_{\ell,\mathrm{screen}} = 4.113\,\mathrm{MPa}$ with the 1.5 section factor and $K_s = 2$. A defensible shear discrepancy cannot be computed without a corresponding FE ligament-shear result. Principal tension is not a substitute; neither it nor these hand values establishes a complete strength pass.

16 / INTERPRETATION

Preserve the deformation relevant to the decision

The broad bridge omits the annulus-to-deck load transfer

The hand bridge assumes the central resultant is already transferred into its broad equivalent section. It does not resolve the local movement needed to transmit load through the narrow connections. This is a plausible contributor to the large discrepancy, consistent with the explicit limitations of the hand model; it is not an isolated or quantified cause.

Subtracting the 44.3 mm slot envelope reduces bridge stiffness compared with an otherwise identical full-width beam. It does not make the surrogate a rigorous upper bound on the actual slotted fixture's motion. A conservative-looking section reduction cannot compensate automatically for an inadequate load-path representation.

Nonuniform annular motion does not isolate ring warping

The annular Y extrema span 0.076435 mm, about 76.44 μm. This establishes spatially nonuniform vertical motion. A rigid surface undergoing rotation can also have nonuniform Y displacement: the extrema alone do not separate local ring warping from translation or rotation. That separation would require a fuller displacement-field assessment or subtraction of a fitted rigid-body motion.

Accordingly, the entire 99.60 μm hand–FE difference cannot be assigned to local ligament or ring flexibility. Geometry, load distribution, support representation and response location all differ. Nor does this range establish rigid-specimen tilt or interface deformation.

The recorded support sequence is not a controlled isolation study

The handoff records an earlier unclamped maximum of 0.13719 mm; the retained bolted, program-controlled screenshot gives 0.13024 mm, **5.07% lower**. The earlier value is a transcription without its original screenshot in this package. Mesh equivalence is not established, so the reduction cannot be attributed exclusively to hold-downs. Both recorded values remain near 0.13 mm.

The later coarse bolted contact-factor change produced a 0.63% decrease (page 12). That particular change had a small effect compared with the hand–FE discrepancy. It does not establish that all numerical contact compliance or support effects were negligible. Compression-only contact also permits separation, unlike the simple-support bridge.

FEA retained important shared and untested assumptions

PLA properties, the 250 N benchmark and annular-only contact remain assumed; agreement cannot validate print architecture, conditioning, creep, failure behavior or the actual force envelope. The model did not test the hand note's 24 mm washer/torque example against real preload. Nor did fixture-only FEA resolve implant/pusher clearance, biological response or complete assembly performance.

> **Assessment of the hand calculations.** They identified the 15 mm web, approximately 6.08 mm connections, shear contribution, local bending and support-span sensitivity. Their local focus was useful for investigating tensile hotspots. The 60 mm bridge was insufficient as a proxy for maximum annular movement: passing its global 50 μm check did not pass the fixture. The original note itself flagged the unresolved local behavior and did not qualify the whole part.

17 / FINAL ANSWER

This PLA concept is not qualified by the present screen

Could printed PLA replace machined metal?

For this geometry and the chosen 250 N / 50 μm screen: no. The final model predicts 131.37 μm maximum downward annular movement, exceeding the target by 81.37 μm. The original hand-only result was insufficient to qualify the slotted part; the geometry-resolved follow-up identifies compliance as the clearest concern under the declared assumptions.

That is not a universal rejection of printed PLA. The actual force envelope, accuracy requirement, print properties, and specimen contact remain unknown. A lower-demand or redesigned printed fixture could be a candidate, but the supplied analysis does not establish its usable load or demonstrate fracture of this one.

Machined 6061 aluminum remains a plausible stiffness alternative. The same-geometry hand model gives about 34 times PLA's flexural stiffness; 304 steel gives about 100 times. These analytical comparisons motivate candidate selection. They do not establish that metal is mandatory, that either material meets the actual test requirement, or that a replacement was manufactured or tested. No metal FE run is included.

What remains necessary before experimental use

1. **Set the requirement:** actual specimen-force envelope, equipment limits, displacement measurement location, and allowable measurement bias.
2. **Verify fit and load transfer:** implant diameter and exit travel, opening clearance, web/ligament dimensions, specimen contact footprint, rigid-base support, and real mounting hardware. A nominal 25 mm implant in a 25 mm opening has zero radial clearance.
3. **Resolve material and local strength:** actual print orientation, walls/infill, conditioning and dwell, relevant properties, and the provisional slot-end stress/mesh limitations. The current elastic FEA supplies no complete failure assessment.
4. **Check the complete assembly physically:** measure fixture motion and recovery under a representative dummy-specimen load within separately established limits. Include base, mounting, pusher, alignment and time-dependent response.

Portfolio wording that preserves the chronology

> In 2025, we set aside this concept because of concerns about PLA breakage. I revisited the supplied CAD in September 2026. At an assumed 250 N, the 60 mm bridge hand calculation predicted 0.03177 mm, 75.82% below the FE model's 0.13137 mm maximum annular movement. The local 5 mm-arm ligament screen gave 5.48 MPa, close to the 5.58/6.05 MPa averaged/unaveraged FE principal peaks, without validating that arm or concentration factor. Final local refinement changed displacement by 0.023% and averaged peak stress by 3.21%. The analysis identified a compliance concern under the chosen assumptions; it did not demonstrate fracture or establish whether the concept met the original experiment's requirements.

Historical boundary: the team feared breakage; no observed breakage is established. Whether this exact concept was printed, and what replacement was ultimately built or tested, remain unverified. Do not attribute the later analysis to the original 2025 decision or claim original authorship of the supplied reference CAD.

18 / EVIDENCE AND REVISION

A traceable record, with limits stated explicitly

What changed across the revisions

Revision A (September 6, 2026) contains the geometry audit and hand calculations. Revision B added the supplied PLA FEA and updated the material conclusion. Revision C (September 7, 2026) adds matched-load displacement and stress comparisons, explicit percentage definitions, the prior 100 mm-span sensitivity, both mesh-refinement steps and a model assessment. It retains the original hand equations and scenario values. No new simulation was performed.

Evidence included with the editable bundle

Record	Location and purpose
Original handoff	`evidence/handoff/`: handoff, model setup, result summary, evidence index, manifest and original verification script, retained unchanged.
Earlier reports	Revision A PDF/source ZIP in `evidence/handoff/sources/`; Revision B PDF, LaTeX and checks in `evidence/revision_b/`.
CAD and screenshots	Original STEP and 15 screenshots in the handoff's `sources/` folder. The two figures reproduced here are the supplied M2 images.
Editable Revision C	`source/`: LaTeX, original analytical script and scenario outputs, CAD figure, and copies of all supplied screenshots.
Revision checks	`source/verify_revision.py` recomputes comparison arithmetic and checks consistency with the preserved handoff; `revision_checks.json` records the output.

Which source supports which claim?

The geometry audit and hand-calculation source support pages 2–8. The recorded setup supports page 9 with the input-audit limitation stated there. M2 annular and principal-stress screenshots support pages 10–11. M0/M1/M2 mesh and result screenshots support page 12; the final plate and washer reaction screenshots support page 13. Pages 14–17 compare those values with the original hand scenarios. The new feedback is retained in `evidence/comparison_feedback.txt`; its interpretation was reviewed against the underlying evidence. The handoff records chronology.

All **27 manifest-listed handoff files passed SHA-256 verification**. All 12 original scenario rows were independently recomputed across 72 numerical output fields with no mismatch. New percentages and equilibrium arithmetic were also checked. These checks establish file integrity and arithmetic consistency, not solver correctness or physical validation.

Coverage and exclusions. No final solver archive/input/results or full solver log was supplied. Some earlier M0/M1 reaction, bearing and M1 quality values exist only as handoff transcriptions. The final balance uses the M2 reaction screenshots. The earlier unclamped value is also a transcription, with mesh equivalence unverified. Missing M2 bearing penetration and final unreported stress components remain unknown, not zero. No real force–displacement measurements, print-specific properties, biological failure model, aluminum simulation, or demonstrated replacement build are established by this record.

The original STEP contains the supplied reference solid, not the later FE face partitions or complete test assembly. Bibliographic statements on the next page retain Revision A's documented source readings; this revision does not create new experimental evidence from literature.

19 / REFERENCES

Literature and source provenance

References

[1] **User-supplied CAD.** `Timmer_NewPushoutforKershLab_v2 v1.step`. Revision A records a single B-rep solid imported using CadQuery/Open CASCADE in millimeters; bounds, face levels, radii, hole positions and slot-end centers were extracted. This revision retains that dimensional audit and CAD-derived figure. This is supplied reference geometry, not an authorship claim.

[2] Dewey, M. J., et al. (2022). *Repair of critical-size porcine craniofacial bone defects using a collagen–polycaprolactone composite biomaterial.* Biofabrication 14, 014102. doi:10.1088/1758-5090/ac30d5. Supplied PDF, pp. 3–4 (article pp. 2–3): specimen geometry and study methods.

[3] Lawson, Z. T., et al. (2021). *Methodology for performing biomechanical push-out tests for evaluating the osseointegration of calvarial defect repair in small animal models.* MethodsX 8, 101541. doi:10.1016/j.mex.2021.101541. Supplied PDF, pp. 3–6 and 10: ABS/infill, clearance, alignment, and the $\leq$ 250 N validation limit.

[4] Dhert, W. J. A., et al. (1992). *A finite element analysis of the push-out test: influence of test conditions.* J. Biomed. Mater. Res. 26, 119–130. doi:10.1002/jbm.820260111. Supplied PDF, pp. 4–6 and 11 (one cover page precedes the article): nominal shear definition, rigid-support assumption, and interpretation limits.

[5] Dewey, M. J., et al. (2019). *Shape-fitting collagen-PLA composite promotes osteogenic differentiation of porcine adipose stem cells.* J. Mech. Behav. Biomed. Mater. 95, 21–33. doi:10.1016/j.jmbbm.2019.03.017. Supplied offline manuscript, Sec. 2.5 / Fig. 4. Distinct initial-fit experiment; no force transferred to the present design requirement.

[6] UltiMaker. *PLA Technical Data Sheet*, pp. 2–3. PDF filename v5.00; internal footer v2.00, April 20, 2022. Printed-coupon properties and conditioning. The lower adopted PLA stiffness/allowables in this note are assumptions, not supplier guarantees.

[7] Kaiser Aluminum. *Rod & Bar Alloy 6061*, KA-RBH-6061-8.10, p. 1. Typical elastic modulus used for the alloy comparison.

[8] Kaiser Aluminum. *Flat Rolled Products*, Table 7.2, KA-SP-AA6-1.09, PDF p. 20. 6061-T651 plate yield limit, 35 ksi, rounded down here to 240 MPa. Purchasing specification and stock certification remain necessary.

[9] Outokumpu. *Core range datasheet*, Tables 5 and 7, pp. 8 and 10. Core 304 plate proof strength on EN basis and elastic modulus. Web references inspected September 6, 2026.

[10] Jerich Lee. *Pigjaw Scaffold–Bone Interface Mechanical Testing* (portfolio), accessed September 6, 2026. Public project framing; not independent verification of experimental outcomes.

[11] **User-supplied analysis handoff** (September 7, 2026). `HANDOFF.md`, `MODEL_SETUP.md`, `results_summary.json`, `EVIDENCE_INDEX.md`, and retained ANSYS screenshots in `evidence/handoff/`. Original project chronology, recorded model setup, M0–M2 results and final scope. Screenshot readouts are not raw solver verification.

Reference links and source page locations are retained from Revision A. Rebuild instructions and complete local evidence paths are in the companion README. The hand calculations and supplied FEA are retrospective numerical screens, not a load rating.